Introduction to Logic (CS) 2026-2027
Homework Assignment 1

Submission deadline: 17th September, 09:00

This homework assignment consists of 6 questions, plus a bonus question. Each question indicates how many points it is worth. With the regular questions, you can earn 100 points. With the bonus question, you can earn additionally 10 points. The grade is the number of points you earned divided by 10, with a maximum of 10.

The submissions instructions are in BrightSpace (Content → Homework Assignments → General Information), **please read them carefully and in full**.

This is an individual assignment. Do not copy your homework from someone (or something) else (this includes LLMs), and do not share your solutions with anyone! If fraud is suspected (e.g., if your answers are too similar compared to the answers of someone else) the Board of Examiners will be notified.

Questions

1. (12 points) For each of the following arguments, say whether it is valid or not valid. Briefly explain your answer. (Note: You are *not* asked to provide a translation into FOL in this question.)

   (a)
   Everybody who submits the project receives feedback.
   Jacob did not receive feedback.
   ∴ Someone did not submit the project.

   (b)
   Every dish that contains garlic is vegetarian.
   The soup is vegetarian.
   ∴ The soup contains garlic.

   (c)
   Every file in the cache is also on the disk.
   There is a file in the cache.
   The user configuration is on the disk.
   Every file in the cache can be read quickly.
   ∴ The user configuration can be read quickly.

2. (4 points) Translate the *propositional logic* formula into natural sounding English:

$$\neg(P \vee Q) \wedge (R \vee S)$$

   Use the following translation key:

   P: Andy is coming to the party

   Q: Kester is coming to the party

   R: There will be cake at the party.

   S: There will be pizza at the party.

3. (21 points) Translate the following sentences into *propositional logic*, where atomic sentences are represented by uppercase letters. Also provide the translation key. Make sure that atomic sentences are translated as full sentences in the translation key. Make sure that your atomic sentences cannot be broken down further.
   (a) Unless I get a deadline extension, I will not go to the party and stay home.
   (b) I will either cycle to work and arrive on time and not be dry, or I will take a bus and not arrive on time but be dry.
   (c) The lunch is either expensive or not warm, but we still eat it every day.

4. (6 points) Translate the following *first-order logic* sentence:

$$\neg P(p(a,b)) \wedge b = p(a,a) \wedge \neg(P(b) \vee L(b,a))$$

   into natural sounding English. The domain of discourse is all natural numbers, and the translation key is as follows:

   constant a: number 2;

   constant b: number 4;

   function $p(x,y)$: the product of x and y;

   unary predicate $P(x)$: x is prime;

   binary predicate $L(x,y)$: x is less than y.

5. (24 points) The following sentences describe scenarios that happened at a chess tournament. Translate the sentences into *first-order logic*, and provide the complete translation key (common to all the sub-questions). You can assume that the domain of discourse consists of all tournament participants.
   (a) Both Kester and Andy played against Eliza, but only one of them beat her.
   (b) Huna's coach is either Kester or Andy, but her coach did not play against Elitsa.
   (c) Andy is his own coach, and he did not beat both Eliza and Elitsa.
   (d) The coach of Kester's coach did not play against Elitsa, and is not Andy.

6. (33 points) Give formal proofs, including justifications, for the following inferences. You can only use the Reiteration rule and the Introduction and Elimination rules for identity and the Boolean connectives.

   (a)
   $\neg P \vee \big(\neg(P \vee Q) \wedge R\big)$
   $\vdash$
   $\neg P$

   (b)
   $R \vee (R \wedge S)$
   $\neg(R \wedge U)$
   $\vdash$
   $R \wedge \neg U$

   (c)
   $\neg R(a,a)$
   $(a = b) \vee (a = c)$
   $R(b,c)$
   $\vdash$
   $\neg(b = c)$

7. (10 points) (Bonus question) Give a formal proof, including justifications, for the following inference. You can only use the Reiteration rule and the Introduction and Elimination rules for identity and the Boolean connectives.

   $\vdash$
   $(P \wedge Q) \vee (\neg P \vee \neg Q)$